CI CD TASK

Question:

CI/CD Pipeline Implementation using Git, Maven, Jenkins, and Docker (Ubuntu)

with provided Maven-based Java web application project.

Tasks

Part A: Project Setup

1. Extract the given project

```
cvr@cm103b-sys07:~/Downloads$ unzip student-webapp.zip
Archive:  student-webapp.zip
  inflating: pom.xml
  inflating: Dockerfile
  inflating: src/main/webapp/index.html
```

2. Verify the project structure

```
cvr@cm103b-sys07:~/Downloads$ ls
Dockerfile          'Experiment 5(1).docx'  'Experiment 7.docx'     'PAGE REPLACEMENT ALGORITHMS.docx'  'Program c(1).docx'  'Shortest Remaining Time First.docx'   student-webapp.zip
'Experiment 3.docx'  'Experiment 5.docx'     'message queues(1).txt'  pipes.txt                          'Program c.docx'     src                                    testcases-banking-lab.xl
'Experiment 4.docx'  'Experiment 6.docx'     'message queues.txt'     pom.xml                            'shared memory.txt'  srtf.txt
```

Part B: Version Control

1. Initialize a Git repository

```
cvr@cm103b-sys07:~/Downloads$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/cvr/Downloads/.git/
```

2. Commit the project files

```
Initialized empty Git repository in /home/cvr/Downloads/student-webapp/.git/
cvr@cm103b-sys07:~/Downloads/student-webapp$ git branch -M main
cvr@cm103b-sys07:~/Downloads/student-webapp$ git add .
```

```
fatal: unable to auto-detect email address (got 'cvr@cm103b-sys07.(none)')
cvr@cm103b-sys07:~/Downloads/student-webapp$ git config --global user.name "dhanasri-4"
git config --global user.email "dhanasri@gmail.com"
cvr@cm103b-sys07:~/Downloads/student-webapp$ git commit -m "Initial commit - Student Web App"
[main (root-commit) b1184b4] Initial commit - Student Web App
 3 files changed, 32 insertions(+)
 create mode 100644 Dockerfile
 create mode 100644 pom.xml
 create mode 100644 src/main/webapp/index.html
```

3. Push the project to GitHub

```
cvr@cm103b-sys07:~/Downloads/student-webapp$ git remote add origin https://github.com/dhanasri-4/project.git
cvr@cm103b-sys07:~/Downloads/student-webapp$ git push -u origin main
Username for 'https://github.com': dhanasri-4
Password for 'https://dhanasri-4@github.com':
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 12 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (8/8), 912 bytes | 912.00 KiB/s, done.
Total 8 (delta 0), reused 0 (delta 0), pack-reused 0
remote: This repository moved. Please use the new location:
remote:   https://github.com/Dhanasri-4/project.git
To https://github.com/dhanasri-4/project.git
 * [new branch]      main -> main
Branch 'main' set up to track remote branch 'main' from 'origin'.
```

Part C: Build Automation

1. Build the project using Maven:

mvn clean package

```
cvr@cm103b-sys07:~/Downloads/student-webapp$ mv src/main/webapp/index_fixed.html src/main/webapp/index.html
cvr@cm103b-sys07:~/Downloads/student-webapp$ mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] ----------------------< com.student:student-webapp >----------------------
[INFO] Building Student Web Application 1.0-SNAPSHOT
[INFO] --------------------------------[ war ]---------------------------------
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ student-webapp ---
[INFO] Deleting /home/cvr/Downloads/student-webapp/target
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ student-webapp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/cvr/Downloads/student-webapp/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.1:compile (default-compile) @ student-webapp ---
[INFO] No sources to compile
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ student-webapp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
```

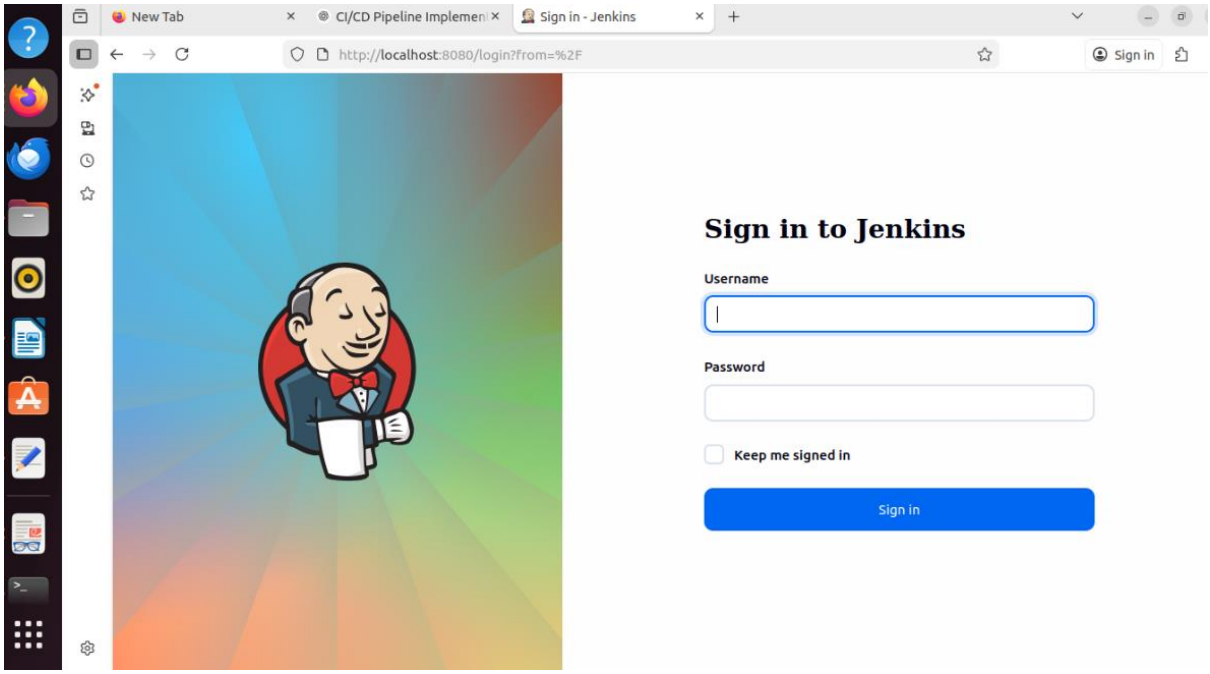
```
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  0.472 s
[INFO] Finished at: 2026-04-06T10:59:29+05:30
[INFO] ------------------------------------------------------------------------
cvr@cm103b-sys07:~/Downloads/student-webapp$ ls -l target/
total 12
drwxrwxr-x 2 cvr cvr 4096 Apr  6 10:59 maven-archiver
drwxrwxr-x 4 cvr cvr 4096 Apr  6 10:59 student-webapp
-rw-rw-r-- 1 cvr cvr 1625 Apr  6 10:59 student-webapp.war
```

2.Verify that a .war file is generated

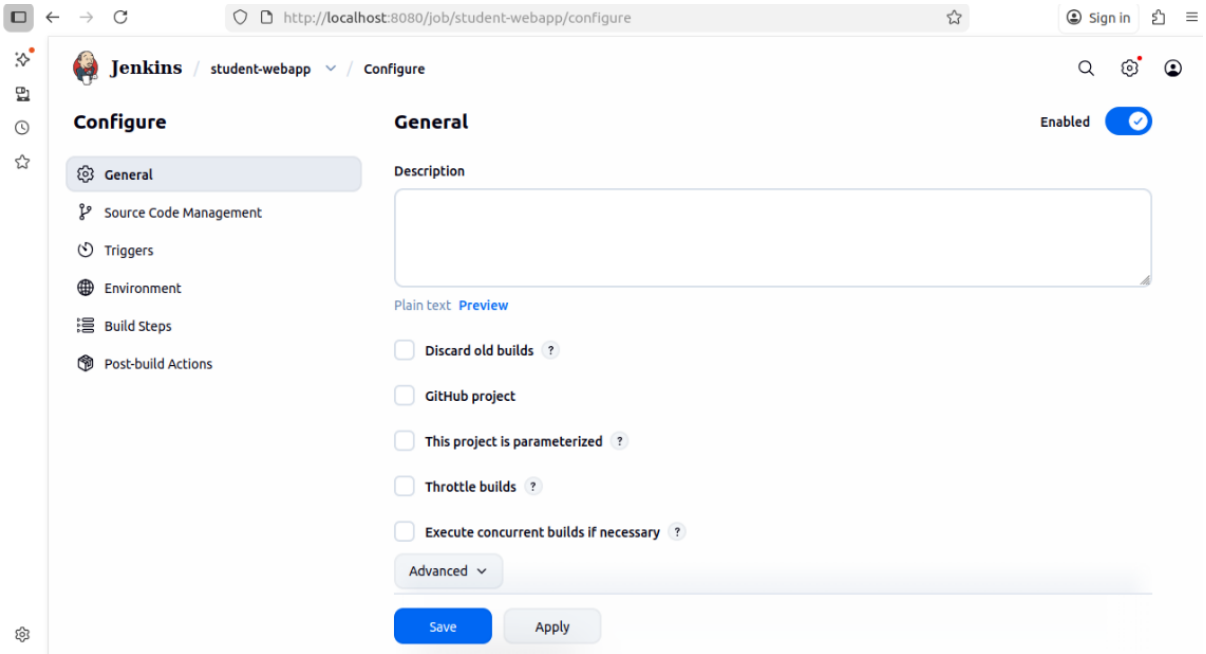
```
cvr@cm103b-sys07:~/Downloads/student-webapp$ ls -l target/
total 12
drwxrwxr-x 2 cvr cvr 4096 Apr  6 10:59 maven-archiver
drwxrwxr-x 4 cvr cvr 4096 Apr  6 10:59 student-webapp
-rw-rw-r-- 1 cvr cvr 1625 Apr  6 10:59 student-webapp.war
```

Part D: Continuous Integration

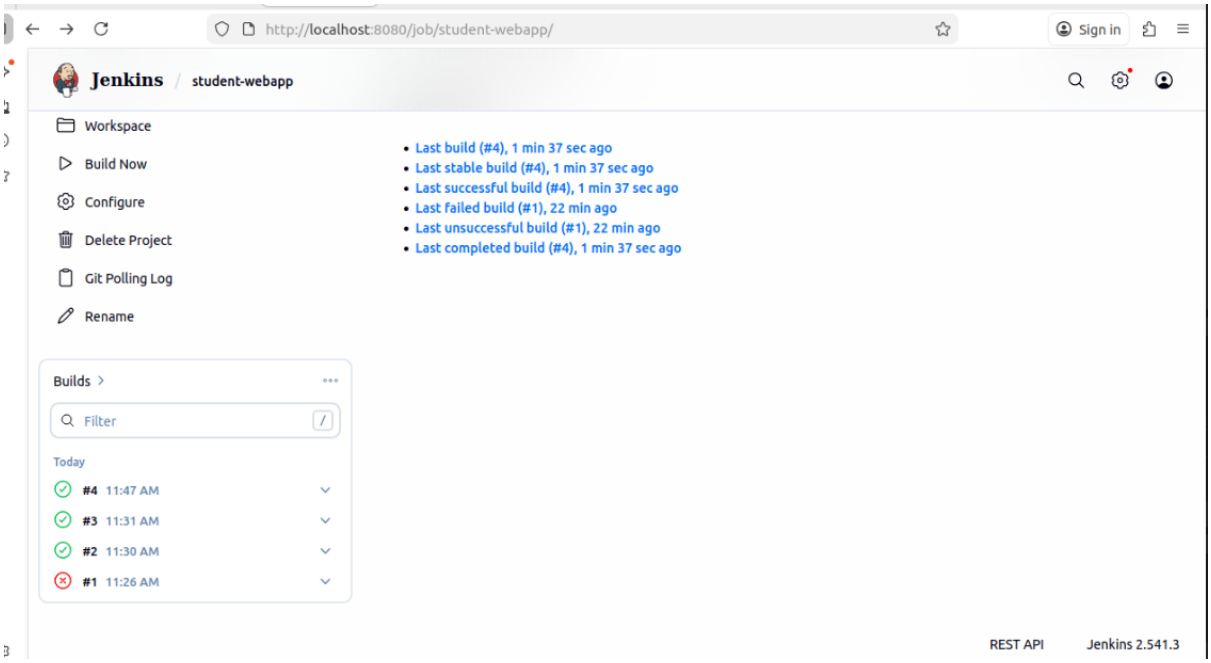
1. Configure Jenkins job



2. Connect to GitHub repository



3. Add build step to run Maven build



Part E: Deployment

1. Use Docker to:

o Build image using Dockerfile

o Deploy .war file into Tomcat container

```
cvr@cm103b-sys07:~/Downloads/student-webapp$ sudo docker ps
CONTAINER ID   IMAGE         COMMAND             CREATED             STATUS              PORTS                                       NAMES
c8f35caf525    student-app   "catalina.sh run"   About a minute ago  Up About a minute   0.0.0.0:8081->8080/tcp, [::]:8081->8080/tcp   adoring_dijkstra
```

.Build Docker Image

```
cvr@cm103b-sys07:~/Downloads/student-webapp$ sudo docker build -t student-app .
[+] Building 34.7s (7/7) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 109B
 => [internal] load metadata for docker.io/library/tomcat:9
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [internal] load build context
 => => transferring context: 1.69kB
 => [1/2] FROM docker.io/library/tomcat:9@sha256:464d89f04614607c27b122c9628f2f154fa514f89496259e12d05234439e2c9f
 => => resolve docker.io/library/tomcat:9@sha256:464d89f04614607c27b122c9628f2f154fa514f89496259e12d05234439e2c9f
 => => sha256:464d89f04614607c27b122c9628f2f154fa514f89496259e12d05234439e2c9f 0.04kB / 0.64kB
 => => sha256:6ee57080dc5b2b6197e7fcd00d1ff04e313a65c02800ef245f58bf1284b4a801 13.12kB / 13.12kB
 => => sha256:c318c44e952a3391b86e5db6a972878c9a4f3f9b51cfbf6d22dcb6d19db6084f 92.39MB / 92.39MB
 => => sha256:945d8a1930d2a7cd0428320257fc3a1872517b0071fe374f4a93ea113e41c00e 2.72kB / 2.72kB
 => => sha256:817887f3c04e0b90b66edc7d903297f121rad2a7c2a3eed5a731557762f91e6c7 29.73MB / 29.73MB
```

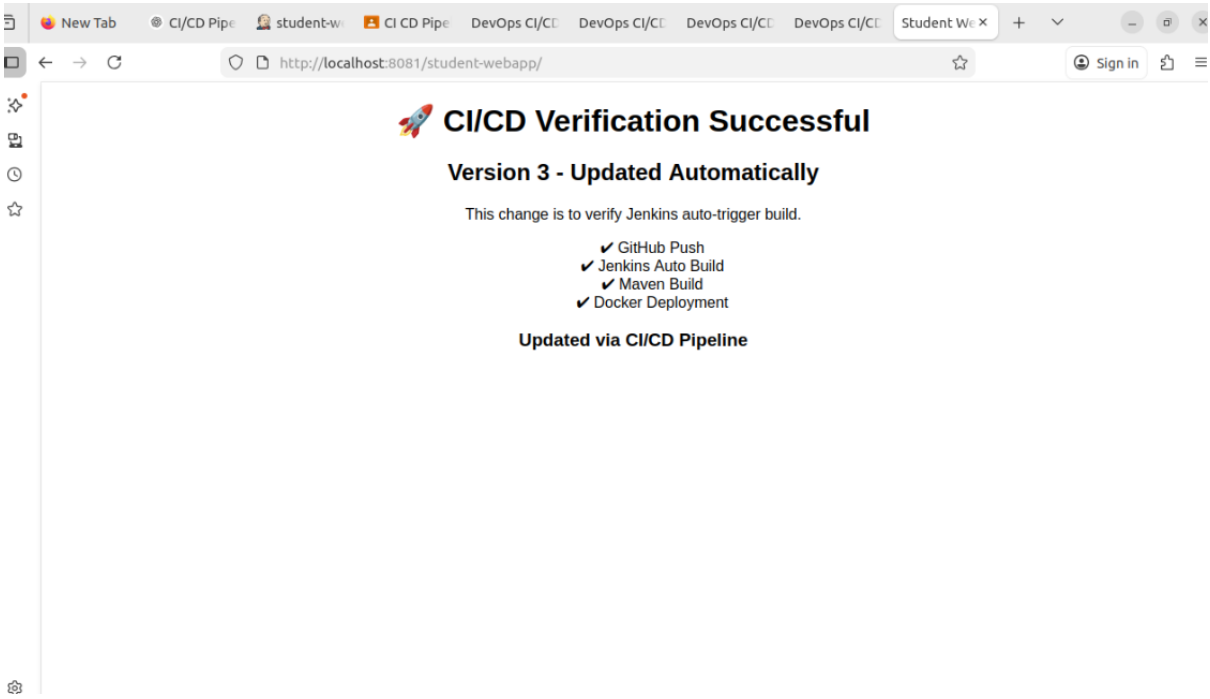
Run Container

```
 => => sha256:cc1e0a39126833701e554d624857517d5d5863a0905ba5539bbe02741c872e51 2.28kB / 2.28kB
 => => sha256:9df0b165ae0b7a998769e983b92cb703c96c3a0217d4504500e2b04be5abfa2e 138B / 138B
 => => sha256:7afe1e4d0f2261617929f52cafc94698f417629a528d7d4ee2b79050ff3add81 15.43MB / 15.43MB
 => => extracting sha256:617887f3c04e0b90b660edc7d903297f121cad2a7c2a3ee05a731557762f91e6c7
 => => extracting sha256:de9be20b95192c91c53b80e11c9949a475df4ba95e71ca9a2ec1da03871891c
 => => extracting sha256:c318c44e952a3391b86e5db6a972878c9a4f3f9b51cfbf6d22dcb6d19db6084f
 => => extracting sha256:4f4fb700ef54461cfa02571ae0dfa0dc1e0cdb55774040d75e08dc38e8acc1
 => => extracting sha256:cc1e0a39126833701e554d624857517d5d5863a0905ba5539bbe02741c872e51
 => => extracting sha256:9df0b165ae0b7a998769e983b92cb703c96c3a0217d4504500e2b04be5abfa2e
 => => extracting sha256:7afe1e4d0f2261617929f52cafc94698f417629a528d7d4ee2b79050ff3add81
 => [2/2] COPY target/student-webapp.war /usr/local/tomcat/webapps/
 => exporting to image
 => => exporting layers
 => => writing image sha256:69492977a4db3cc819edd858848861f4cb3698f7c6f88f38e09d79b3730935eb
 => => naming to docker.io/library/student-app
cvr@cm103b-sys07:~/Downloads/student-webapp$ sudo docker run -d -p 8081:8080 student-app
2113a1bc3cc23a5e56f704059f9df2165daeee9a0c72c672807ddf98fd4df855
```

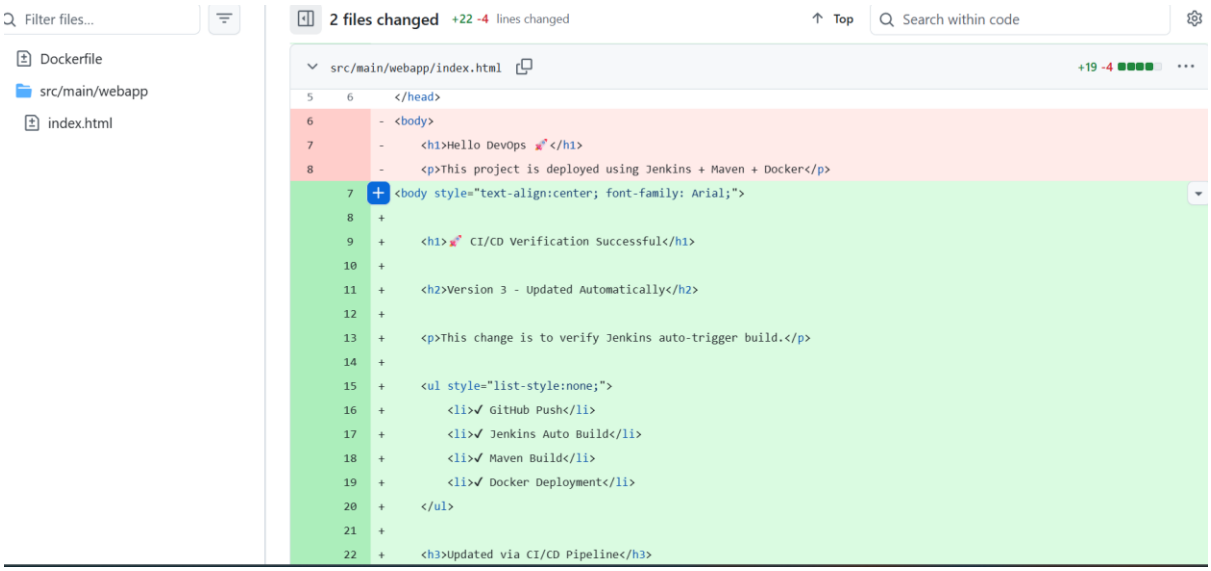
```
cvr@cm103b-sys07:~/Downloads/student-webapp$ cp target/your-app.war /path/to/tomcat/webapps/
```

Part F: CI/CD Verification

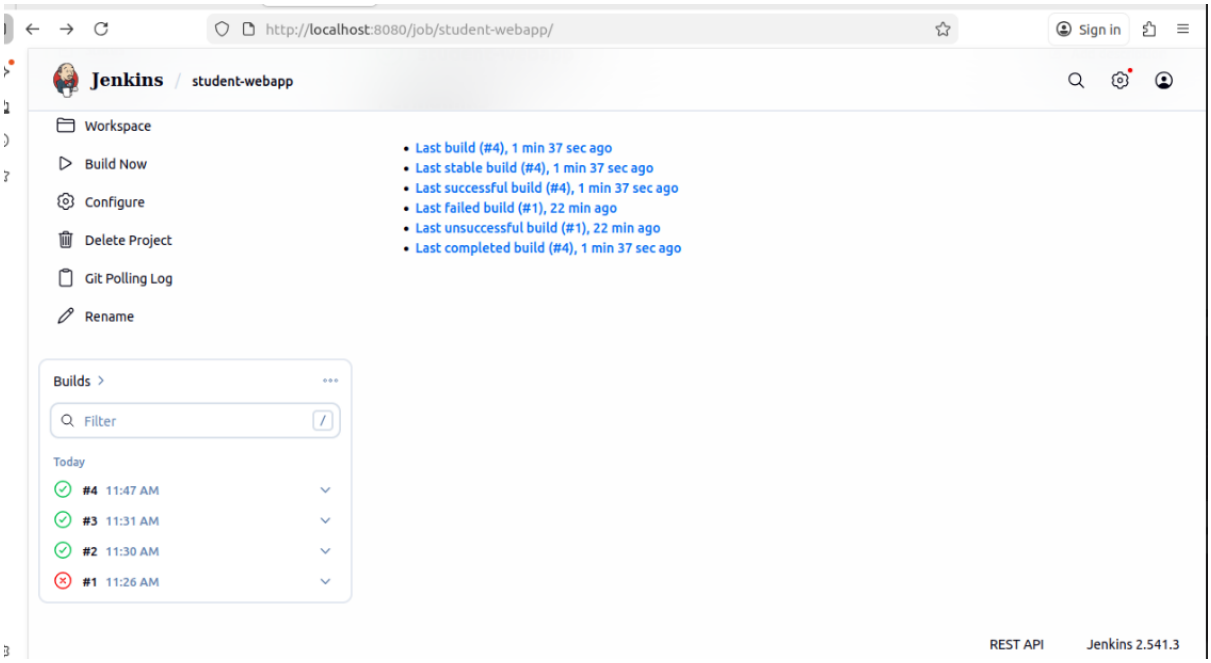
1. Modify index.html



2. Push changes



o Jenkins auto-trigger


http://localhost:8080/job/student-webapp/
Sign in
Jenkins
student-webapp
Workspace
Build Now
Configure
Delete Project
Git Polling Log
Rename
Last build (#4), 1 min 37 sec ago
Last stable build (#4), 1 min 37 sec ago
Last successful build (#4), 1 min 37 sec ago
Last failed build (#1), 22 min ago
Last unsuccessful build (#1), 22 min ago
Last completed build (#4), 1 min 37 sec ago
Builds
Filter
Today
#4 11:47 AM
#3 11:31 AM
#2 11:30 AM
#1 11:26 AM
REST API
Jenkins 2.541.3